



Technical Specifications:

- Wingspan: 116cm (45.7 in)
- Weight (incl. camera and battery): 1.3kg-1.6kg depending on camera and battery
- Electric motor: Low-noise, brushless
- Radio link range: 3km nominal (up to 8km)/1.9 mi (up to 5 mi)
- Detachable wings: Yes
- Cameras (optional): senseFly S.O.D.A. 3D, senseFly Aerial X, senseFly Duet T, Parrot Sequoia+, senseFly S.O.D.A., senseFly Corridor, MicaSense RedEdge MX
- GNSS grade: Survey
- Carry backpack dimensions: 75×50×29cm
- Backpack weight (empty): 4.6kg
- Carry case dimensions: 75×48×33cm

Fig. 3: Sensefly eBee X

during day as well as night-time. In addition, they use motion sensors and accelerometers for calculating and recording the vehicle movement and even the direction of wildlife animals and/or birds.

Modern AI solutions like footprint identification technique (FIT) are being experimented with for wildlife population tracking and census, animal movement and migration in different seasons, and any occasional or regular seasonal health issues or threats to wildlife animals and birds. The recorded data can be stored in a data pool and used for analytics such as historical data evaluation, any regional and climatic understanding of specific and rare wildlife, and environmental changes over a period of time to study their eco-system.

Following are some of the best wildlife monitoring drones known at present:

Super Bat DA-50. Powered by batteries, its flight time ranges from eight to ten hours. It has a 2-stroke gasoline engine for flight efficiency and longevity. Its operational flight altitude peaks at about 4,600 metres

(15,000 feet) at 72.5 to 129 kilometres an hour. Plus, its antenna can send information to the system's ground station along with receiving and sending telemetry.

Air Shepherd ZT-TIC. It was developed specifically for anti-poaching and is in use in several parks in Africa. With a service ceiling of about 1,220 metres (4,000 feet) and an endurance of around five hours, it includes a day and a night camera capable of sending live video feed to an operator up to 30km away. Can fly an automated route and has a control range of 50km line-of-site.

The long-endurance, high-altitude powerful camera and range make the Super Bat and Air Shepherd ideal for covering large areas that encompass most refuges. Both cover ground faster than anyone on foot or by ground vehicle that could alert poachers about rangers looking to catch them. Typical employment of these drones is to determine, based on past patterns, where poachers are most likely to enter the park and operate.

Sensefly eBee X. This is one of the first kind of a fixed-wing mapping and surveying drone, which

is usually light-weight and can be launched from hand. It is used by forest rangers, surveyors, and geo-spatial professionals to manage, monitor, and collect various aerial mapping data, location details, and environmental conditions using built-in sensors and camera.

The raw data collected by Sensefly can be used to maintain and dynamically create stunning high-resolution orthomosaic images. Or it can be used by photogrammetry software, such as Pix4Dmapper, to design and create topographic maps, including DSMs, DEMs, 3D points clouds, and more. We can get critical project insights from these topographic data, which can be shared with wildlife researchers, bird photographers, and animal healthcare professionals for better decision-making.

eBee X is also available with RTK or PPK for greater positional accuracy.

The future of drones isn't in the skies. It's in the ocean. "It feels like an amusement park ride—except it's real. You're really driving around underwater, exploring."—Gizmodo

OpenROV. It is a remotely operated 15cm x 20cm x 30cm mini-submarine that weighs 2.6kg. This submarine is powered by eight 26650-format Li-ion batteries and can be assembled from common items, with the most expensive being Beaglebone Black Linux computer. The submarine is controlled from a laptop computer connected to the submarine via a tether and is equipped with on-board LEDs and a camera. OpenROV is an open source hardware project. Website: <http://openrov.com>; Latest version: 30.1.0

Wildlife surveillance and counter poaching

Constant wildlife surveillance is always useful to avoid man-animal clashes when they cross their paths. There are many wild animals that have been categorised as rare